

Chapter 1

Errors in Numerical Computations

The solutions of mathematical problems are of two types: analytical and numerical. The analytical solutions can be expressed in closed form and these solutions are error free. On the other hand, numerical method is a division of mathematics which solves problems using computational machine (computer, calculator, etc.). But, for some classes of problems it is very difficult to obtain an analytical solution. For example, the Indian populations are known at the years 1951, 1961, 1971, 1981, 1991, 2001. There is no analytical method available to determine the population in the year, say, 2000. But, using numerical method one can determine the population in the said year. Again, sometimes we observed that the solutions of non-linear differential equations cannot be determined by analytical methods, but, such problems can easily be solved by numerical methods. Numerical computations are almost invariably contaminated by errors, and it is important to understand the source, propagation, magnitude, and rate of growth of these errors.

In this age of computer, many complicated and large problems are solved in significantly less time. But, without using numerical methods we cannot solve any mathematical problem using computer, as analytical methods are not suitable to solve a problem by computer. Thus, the numerical methods are highly appreciated and extensively used by Mathematicians, Computer Scientists, Statisticians, Engineers and others.

1.1 Sources of Errors

The solution of a problem obtained by numerical method contains some errors. To minimize the errors, it is most essential to identify the causes or sources of the errors

and their growth and propagation in numerical computation. Three types of errors, viz., inherent errors, round-off errors and truncation errors, occur in finding the solution of a problem using numerical method. These three type of errors are discussed below.

(i) **Inherent errors:** This type of errors is present in the statement of the problem itself, before determining its solution. Inherent errors occur due to the simplified assumptions made in the process of mathematical modelling of a problem. It can also arise when the data is obtained from certain physical measurements of the parameters of the proposed problem.

(ii) **Round-off errors:** Generally, the numerical methods are carried out using calculator or computer. In numerical computation, all the numbers are represented by decimal fraction. Some numbers such as $1/3$, $2/3$, $1/7$ etc. can not be represented by decimal fraction in finite numbers of digits. Thus, to get the result, the numbers should be rounded-off into some finite number of digits.

Again, most of the numerical computations are carried out using calculator and computer. These machines can store the numbers up to some finite number of digits. So in arithmetic computation, some errors will occur due to the finite representation of the numbers; these errors are called round-off error. Thus, round-off errors occur due to the finite representation of numbers during arithmetic computation. These errors depend on the word length of the computational machine.

(iii) **Truncation errors:** These errors occur due to the finite representation of an inherently infinite process. For example, the use of a finite number of terms in the infinite series to compute the value of $\cos x$, $\sin x$, e^x , etc.

The Taylor's series expansion of $\sin x$ is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

This is an infinite series expansion. If only first five terms are taken to compute the value of $\sin x$ for a given x , then we obtain an approximate result. Here, the error occurs due to the truncation of the series. Suppose, we retain the first n terms, the truncation error (E_{trunc}) is given by

$$E_{\text{trunc}} \leq \frac{x^{2n+1}}{(2n+1)!}.$$

It may be noted that the truncation error is independent of the computational machine.

1.2 Exact and Approximate Numbers

To solve a problem, two types of numbers are used. They are exact and approximate. Exact number gives a true value of a result and approximate number gives a value which is closed to the true value.

For example, in the statements ‘a triangle has three sides’, ‘there are 2000 people in a locality’, ‘a book has 450 pages’ the numbers 3, 2000 and 450 are exact numbers. But, in the assertions ‘the height of a pupil is 178 cm’, ‘the radius of the Earth is 6400 km’, ‘the mass of a match box is ten gram’, the numbers 178, 6400 and 10 are approximate numbers.

This is due to the imperfection of measuring instruments we use. There are no absolutely exact measuring instruments; each of them has its own accuracy. Thus, the height of a pupil is 178 cm is not absolute measurement. In the second example, the radius of the Earth is very concept; actually, the Earth is not a sphere at all, and we can use its radius only in approximate terms. In the last example, the approximation of the number is also defined by the fact that different boxes may have different masses and the number 10 defines the mass of a particular box.

One important observation is that, same number may be exact as well as approximate. For example, the number 3 is exact when it represents the number of sides of a triangle and approximate if we use it to represent the number π when calculating the area of a circle using the formula πr^2 .

Independently, the numbers 1, 2, 3, $\frac{1}{2}$, $\frac{5}{3}$, $\sqrt{2}$, π , e , etc. written in this manner are exact. An approximate value of π is 3.1416, a better approximation of it is 3.14159265. But one cannot write the exact value of π .

The accuracy of calculations is defined by the number of digits in the result which enjoy confidence. The **significant digits** or **significant figures** of a number are all its digits, except for zeros which appear to the left of the first non-zero digit. Zeros at the end of a number are always significant digit. For example, the numbers 0.001205 and 356.800 have 4 and 6 significant digits respectively.

In practical calculations, some numbers occur containing large number of digits, and it will be necessary to cut them to a usable number of figures. This process is called **rounding-off** of numbers. That is, in rounding process the number is replaced by another number consisting of a smaller number of digits. In that case, one or several digits keep with the number, taken from left to right, and discard all others.

The following rules of rounding-off are commonly used:

- (i) *If the discarded digits constitute a number which is larger than half the unit in the last decimal place that remains, then the last digit that is left is increased by one.*

If the discarded digits constitute a number which is smaller than half the unit in the last decimal place that remains, then the digits that remain do not change.

- (ii) *If the discarded digits constitute a number which is equal to half the unit in the last decimal place that remains, then the last digit that is half is increased by one, if it is odd, and is unchanged if it is even.*

This rule is often called a rule of an **even digit**. If a number is rounded using the above rule then the number is called correct up to some (say n) significant figures.

The following numbers are rounded-off correctly to *five* significant figures:

Exact number	Round-off number
25.367 <u>8</u> 35	25.368
28.353 <u>2</u> 15	28.353
3.7853 <u>5</u> 3	3.7854
5.8354 <u>5</u> 3	5.8355
6.7354 <u>5</u>	6.7354
4.8327 <u>5</u>	4.8328
0.0058345 <u>7</u> 8	0.0058346
38567 <u>5</u> 4	38568×10^2
2.37	2.3700
8.9999 <u>7</u>	9.0000
9.9999 <u>8</u>	10.000

From above examples, it is easy to observe that, while rounding a number, an error is generated and this error is sometimes called **round-off** error.

1.3 Absolute, Relative and Percentage Errors

Let x_T be the exact value of a number and x_A be its approximate value. If $x_A < x_T$, then we say that the number x_A is an approximate value of the number x_T by **defect** and if $x_A > x_T$, then it is an approximate value of x_T by **excess**.

The difference between the exact value x_T and its approximate value x_A is an **error**. As a rule, it is not possible to determine the value of the error $x_T - x_A$ and even its sign, since the exact number x_T is unknown.

The errors are represented in three ways, viz., absolute error, relative error and percentage error.

Absolute error:

The absolute error of the approximate number x_A is a quantity (Δx) which satisfies the inequality

$$\Delta x \geq |x_T - x_A|.$$

The absolute error is the upper bound of the deviation of the exact number x_T from its approximation, i.e.,

$$x_A - \Delta x \leq x_T \leq x_A + \Delta x.$$

The above result can be written in the form

$$x_T = x_A \pm \Delta x. \quad (1.1)$$

In other words, the absolute error of the number x is the difference between true value and approximate value, i.e.,

$$\Delta x = |x_T - x_A|.$$

It may be noted from the rounding process that, if a number be rounded to m decimal places then

$$\text{absolute error} \leq \frac{1}{2} \times 10^{-m}. \quad (1.2)$$

The absolute error measures only the **quantitative** aspect of the error but not the qualitative one, i.e., does not show whether the measurement and calculation were accurate. For example, the length and the width of a table are measured with a scale (whose division is 1 cm) and the following results are obtained: the width $w = 5 \pm 0.5$ cm and the length $l = 100 \pm 0.5$ cm. In both cases the absolute error is same and it is 0.5 cm. It is obvious that the second measurement was more accurate than the first. To estimate the quality of calculations or measurements, the concept of a relative error is introduced.

Relative error:

The relative error (δx) of the number x_A is

$$\delta x = \frac{\Delta x}{|x_A|} \text{ or } \frac{\Delta x}{|x_T|}, \quad |x_T| \neq 0 \text{ and } |x_A| \neq 0.$$

This expression can be written as

$$x_T = x_A(1 \pm \delta x) \text{ or } x_A = x_T(1 \pm \delta x).$$

Note that relative error is the absolute error when measuring 1 unit.

For the measurements of the length and the width of the table (discussed earlier) the relative errors are

$$\delta w = \frac{0.5}{5} = 0.1 \text{ and } \delta l = \frac{0.5}{100} = 0.005.$$

In these cases, one can conclude that the measurement of the length of the table has been relatively more accurate than that of its width. So one conclusion can be drawn: *the relative error measures the quantity and quality of the calculation and measurement. Thus, the relative error is a better measurement of error than absolute error.*

Percentage error:

The percentage error of an approximate number x_A is $\delta x \times 100\%$.

It is a particular type of relative error. This error is sometimes called **relative percentage error**. The percentage error gives the total error while measuring 100 unit instead of 1 unit. This error also calculates the quantity and quality of measurement. When relative error is very small then the percentage error is calculated.

Note 1.3.1 The absolute error of a number correct to n significant figures cannot be greater than half a unit in the n th place.

Note 1.3.2 The relative error and percentage error are independent of the unit of measurement, while absolute error depends on the measuring unit.

Difference between relative error and absolute error:

Absolute error measures only quantity of error and it is the total amount of error incurred by approximate value. While the relative error measures both the quantity and quality of the measurement. It is the total error while measuring one unit. The absolute error depends on the measuring unit, but, relative error does not depend on measuring unit.

Example 1.3.1 Find the absolute, relative and percentage error in x_A when $x_T = \frac{1}{3}$ and $x_A = 0.333$.

Solution. The absolute error

$$\begin{aligned}\Delta x &= |x_T - x_A| = \frac{1}{3} - 0.333 = \frac{1 - 0.999}{3} \\ &= \frac{0.001}{3} = 0.00033.\end{aligned}$$

The relative error

$$\delta x = \frac{\Delta x}{x_T} = \frac{0.00033}{1/3} = 0.00099 \simeq 0.001.$$

The percentage error is $\delta x \times 100\% = 0.00099 \times 100\% = 0.099\% \simeq 0.1\%$.

Example 1.3.2 An exact number x_T is in the interval $[28.03, 28.08]$. Assuming an approximate value, find the absolute and the percentage errors.

Solution. The middle of the given interval is taken as its approximate value, i.e., $x_A = 28.055$. The absolute error is half of its length, i.e., $\Delta x = 0.025$. The relative error $\delta x = \frac{\Delta x}{x_A} = 0.000891 \dots$.

It is conventional to round-off the error to one or two non-zero digits. Therefore, $\delta x = 0.0009$ and the percentage error is 0.09% .

Example 1.3.3 Determine the absolute error and the exact number corresponding to the approximate number $x_A = 5.373$ if percentage error is 0.01%.

Solution. Here the relative error $\delta x = 0.01\% = 0.0001$.
 The absolute error $\Delta x = |x_A| \times \delta x = 5.373 \times 0.0001 = 0.0005373 \simeq 0.00054$.
 The exact value $= 5.373 \pm 0.00054$.

Example 1.3.4 Find out in which of the following cases, the quality of calculations is better: $x_T = \frac{15}{17} \simeq 0.8824$ and $y_T = \sqrt{51} \simeq 7.141$.

Solution. To find the absolute error, we take the numbers x_A and y_A with a larger number of decimal digits as $x_A \simeq 0.882353$, $y_A = \sqrt{51} \simeq 7.141428$.
 Therefore, the absolute error in x_T is $|0.882353 \dots - 0.8824| \simeq 0.000047$,
 and in y_T is $|7.141428 \dots - 7.141| \simeq 0.00043$.
 The relative error in x_A is $0.00047/0.8824 \simeq 0.00053 = 0.05\%$
 and relative error in y_A is $0.00043/7.141 = 0.0000602 = 0.006\%$.

In the second case the quality of calculation is better than the first case as relative error in $x_T >$ relative error in y_T .

1.4 Valid Significant Digits

A real number can be represented by many different ways. For example, the number 840000 can be represented as two factors: 840×10^3 or 84.0×10^4 or 0.840×10^6 . (Note that in these representations the last three significant zeros are lost). The later form of the notation is known as **normalize form** and it is commonly used. In this case, we say that 840 is the mantissa of the number and 6 is its **order**.

Every positive decimal number, exact as well as approximate, can be expressed as

$$a = d_1 \times 10^m + d_2 \times 10^{m-1} + \dots + d_n \times 10^{m-n+1} + \dots,$$

where d_i are the digits constituting the number ($i = 1, 2, \dots$) with $d_1 \neq 0$ and 10^{m-i+1} is the value of the i th decimal position (counting from left).

The digit d_n of the approximate number a is **valid significant digit** (or simply a **valid digit**) if it satisfies the following inequality.

$$\Delta a \leq 0.5 \times 10^{m-n+1}, \quad (1.3)$$

i.e., absolute error does not exceed half the unit of the decimal digit in which d_n appears.

If inequality (1.3) is not satisfied, then the digit d_n is said to be **doubtful**. It is obvious that if the digit d_n is valid, then all the preceding digits, to the left of it, are also valid.

Theorem 1.1 *If a number is correct up to n significant figures and the first significant digit of the number is k , then the relative error is less than*

$$\frac{1}{k \times 10^{n-1}}.$$

Proof. Let x_A be the approximate value of the exact number x_T . Also, let x_A is correct up to n significant figures and m decimal places. Then there are three possibilities may occur:

- (i) $m < n$
- (ii) $m = n$ and
- (iii) $m > n$.

We have by (1.2), the absolute error $\Delta x \leq 0.5 \times 10^{-m}$.

Case I. When $m < n$.

In this case, the total number of digits in integral part is $n - m$. If k be the first significant digit in x_T , then

$$\Delta x \leq 0.5 \times 10^{-m} \text{ and } |x_T| \geq k \times 10^{n-m-1} - 0.5 \times 10^{-m}.$$

Therefore, the relative error

$$\begin{aligned} \delta x &= \frac{\Delta x}{|x_T|} \leq \frac{0.5 \times 10^{-m}}{k \times 10^{n-m-1} - 0.5 \times 10^{-m}} \\ &= \frac{1}{2k \times 10^{n-1} - 1}. \end{aligned}$$

Since, n is a positive integer and k is an integer lies between 1 and 9,

$$2k \times 10^{n-1} - 1 > k \times 10^{n-1}$$

for all k and n except $k = n = 1$.

Hence,

$$\delta x < \frac{1}{k \times 10^{n-1}}.$$

Case II. When $m = n$.

In this case, the first significant digit is the first digit after decimal point, i.e., the integral part is zero.

As before,

$$\begin{aligned} \delta x &= \frac{0.5 \times 10^{-m}}{k \times 10^{n-m-1} - 0.5 \times 10^{-m}} \\ &= \frac{1}{2k \times 10^{n-1} - 1} < \frac{1}{k \times 10^{n-1}}. \end{aligned}$$

Case III. When $m > n$.

In this case, the first significant digit k is at the $(n - m + 1) = -(m - n - 1)$ th position. Also, the integer part is zero. Then $\Delta x \leq 0.5 \times 10^{-m}$ and $|x_T| \geq k \times 10^{-(m-n+1)} - 0.5 \times 10^{-m}$.

Therefore,

$$\begin{aligned} \delta x &= \frac{0.5 \times 10^{-m}}{k \times 10^{-(m-n+1)} - 0.5 \times 10^{-m}} \\ &= \frac{1}{2k \times 10^{n-1} - 1} < \frac{1}{k \times 10^{n-1}}. \end{aligned}$$

Hence the theorem.

1.5 Propagation of Errors in Arithmetic Operations

1.5.1 The errors in sum and difference

Consider the exact numbers $X_1, X_2, \dots, X_n$ and their approximations be respectively $x_1, x_2, \dots, x_n$. Let $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ be the errors in $x_1, x_2, \dots, x_n$, i.e., $X_i = x_i \pm \Delta x_i, i = 1, 2, \dots, n$. Also, let $X = X_1 + X_2 + \dots + X_n$ and $x = x_1 + x_2 + \dots + x_n$.

Therefore, the total absolute error is

$$\begin{aligned} |X - x| &= |(X_1 - x_1) + (X_2 - x_2) + \dots + (X_n - x_n)| \\ &\leq |X_1 - x_1| + |X_2 - x_2| + \dots + |X_n - x_n|. \end{aligned}$$

Thus the absolute error in the sum is

$$|\Delta x| = |\Delta x_1| + |\Delta x_2| + \dots + |\Delta x_n|. \quad (1.4)$$

Thus the absolute error in sum of approximate numbers is equal to the sum of the absolute errors of the numbers.

From (1.4), it follows that the absolute error of the algebraic sum must not be smaller than the absolute error of the least exact term.

The following points should be kept in mind when adding numbers of different absolute accuracy.

- (i) identify a number (or numbers) of the least accuracy (i.e., a number which has the maximum absolute error),
- (ii) round-off more exact numbers so as to retain in them one digit more than in the identified number (i.e., retain one reserve digit),
- (iii) perform addition taking into account all the retained digits,
- (iv) round-off the result by discarding one digit.